BUSINESS CASE CHAMPIONSHIP 2026 *CASE #9 - AGROBOT DELIVERY*

ARCUS

Commercialization & market-entry strategy for EFKO's **humanoid delivery robot**

Russia, UAE, India

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The last mile is **breaking** - and EFKO built a robot for it

50–60%

of logistics cost is the last mile

300,000

couriers Moscow needs by 2030

+15% / yr

courier wages, 35–45% turnover

Wheeled robots stop at the building. Arcus opens doors, takes lifts and reaches the apartment - **the “last 20 metres”.**

Not a robot - a **delivery system**

Humanoid courier	opens doors, uses lifts, climbs steps
Roboports	autonomous battery-swap & loading stations
Postamats	automated parcel lockers
≤ 1 hour	access to 20,000 SKU , up to 4× cheaper delivery

Honest today: a prototype, operator-controlled. The strategy is built on logic, not hype.

Where to launch? How to sell? With whom?

3 MARKETS

Russia, UAE, India

3 MODELS

Internal, B2B sale, **RaaS**

FRAMEWORKS

PESTLE , Porter’s Five Forces , Gartner Hype Cycle , Risk Matrix , Triple Bottom Line , Hofstede , Modes of Entry , Ansoff , Ghemawat AAA , Benchmarking , Value Proposition Canvas , AMC , Lean/Agile , SMART

1 UAE leads the macro-environment

PESTLE: tailwind in the Gulf, barrier in India, mixed in Russia

2 The humanoid door-to-door niche is empty

Porter: rivalry low (2/5); supplier & buyer power high (4/5)

3 Ahead on concept, early on maturity

Gartner Hype Cycle: discipline over hype - an evidence-gated pilot

UAE a tailwind, Russia mixed, **India a barrier**

	RUSSIA	UAE	INDIA
Political	+1 State-backed; sanctions on parts	+2 Top-down; Dubai AV Strategy 2030	−1 Fragmented; weak central push
Economic	+2 Last mile 42% of cost; wages +15%/yr	+1 GDP/cap $43k; order value $45	−1 GDP/cap $2.4k; labour $1.2/hr
Social	−2 Willingness 28%; job-fear 72%	+2 Willingness 52%; youth 75% positive	−2 Willingness 22%; job-fear 82%
Technological	+1 Infra 55; broken lifts = last 20 m	+1 Infra 78; smart buildings 88%	−2 Infra 32; lifts ~25%
Legal	−2 Draft law; approval 36–60 mo	+2 Pilots allowed; approval 12–24 mo	−2 Undefined; approval 48–84 mo
Environmental	−1 Cold, snow, ice	−1 Extreme heat 50 °C	−1 Monsoon + heat

Three forces press hard — but the niche is **empty**

Porter's Five Forces — assessed intensity (1 = low, 5 = high)



Source: case data & competitive landscape (report Table 2).

ASSESSED INTENSITY (1 = LOW … 5 = HIGH)

Supplier power **4 · High**
Actuators ≈80% of cost; sanctions on Western parts

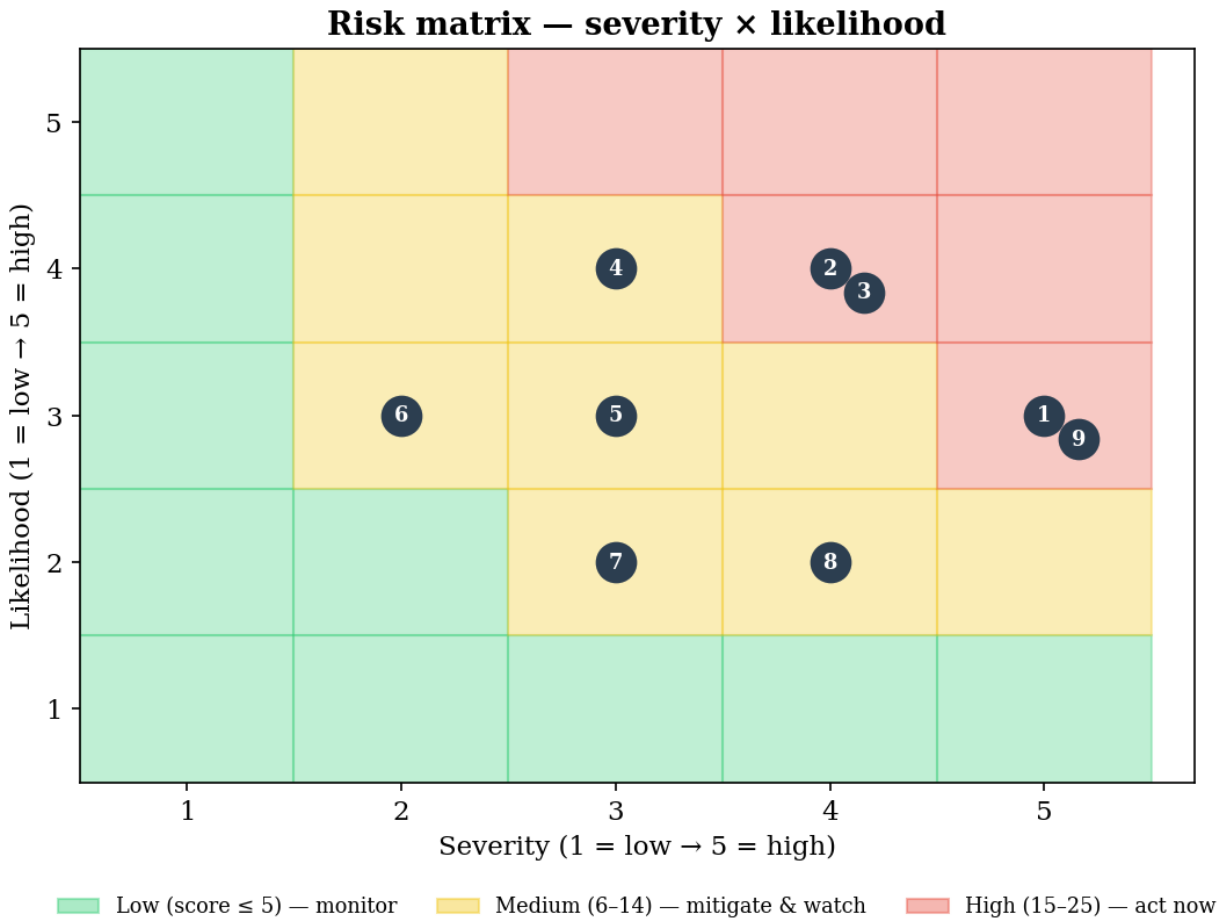
Buyer power **4 · High**
Retailers compare vs $1.8–2.5 rovers, $3–12.5 couriers

Threat of substitutes **4 · High**
Couriers, drones, parcel lockers, dark-store pickup

Threat of new entrants **3 · Moderate**
High capex / IP / regulation, but funded startups

Competitive rivalry **2 · Low**
Humanoid door-to-door niche is essentially empty

Rivalry is the opening: Arcus enters an unoccupied niche, shielded from a direct clash while supplier and buyer power are managed by ~70% localisation and the RaaS model.

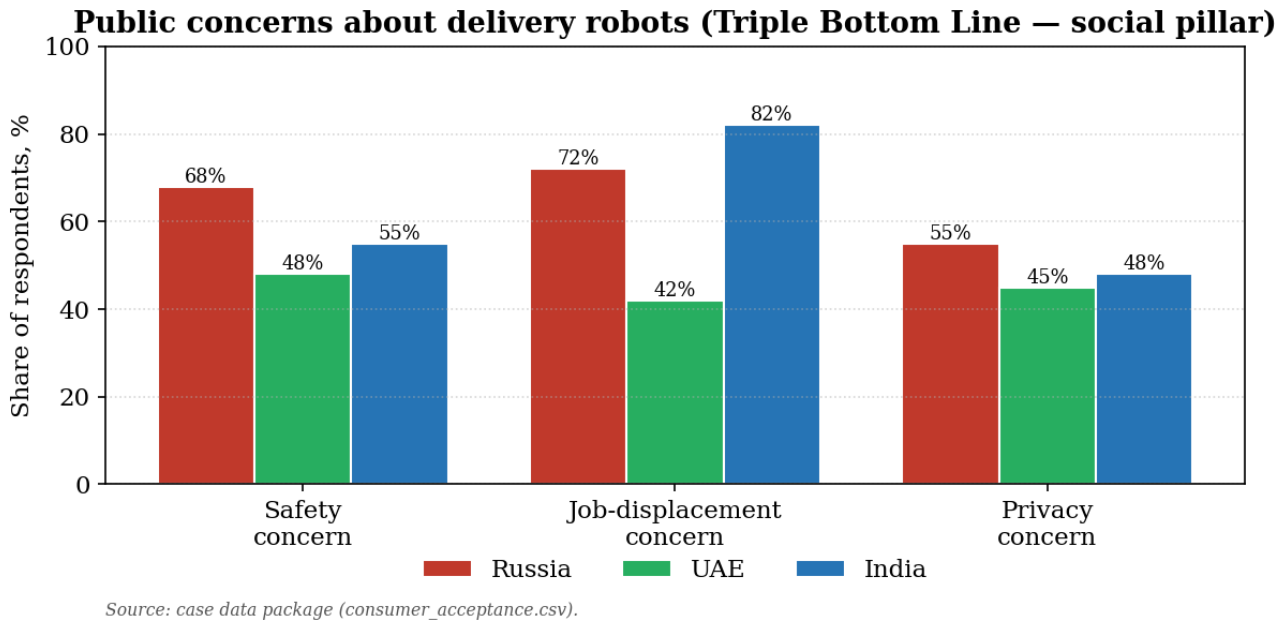


4 of 9 risks are red-zone

No anthropomorphic-robot law (RU)	**16**
Hand / actuator reliability	**16**
Public safety incident	**15**
Unit economics don’t close	**15**

Job-loss fear 72% (RU) / 82% (India). Answer: augmentation, not replacement - reskilling amid a real labour shortage.

The sharpest barrier is social — met with **“augmentation, not replacement”**



Source: case data package (consumer_acceptance.csv).

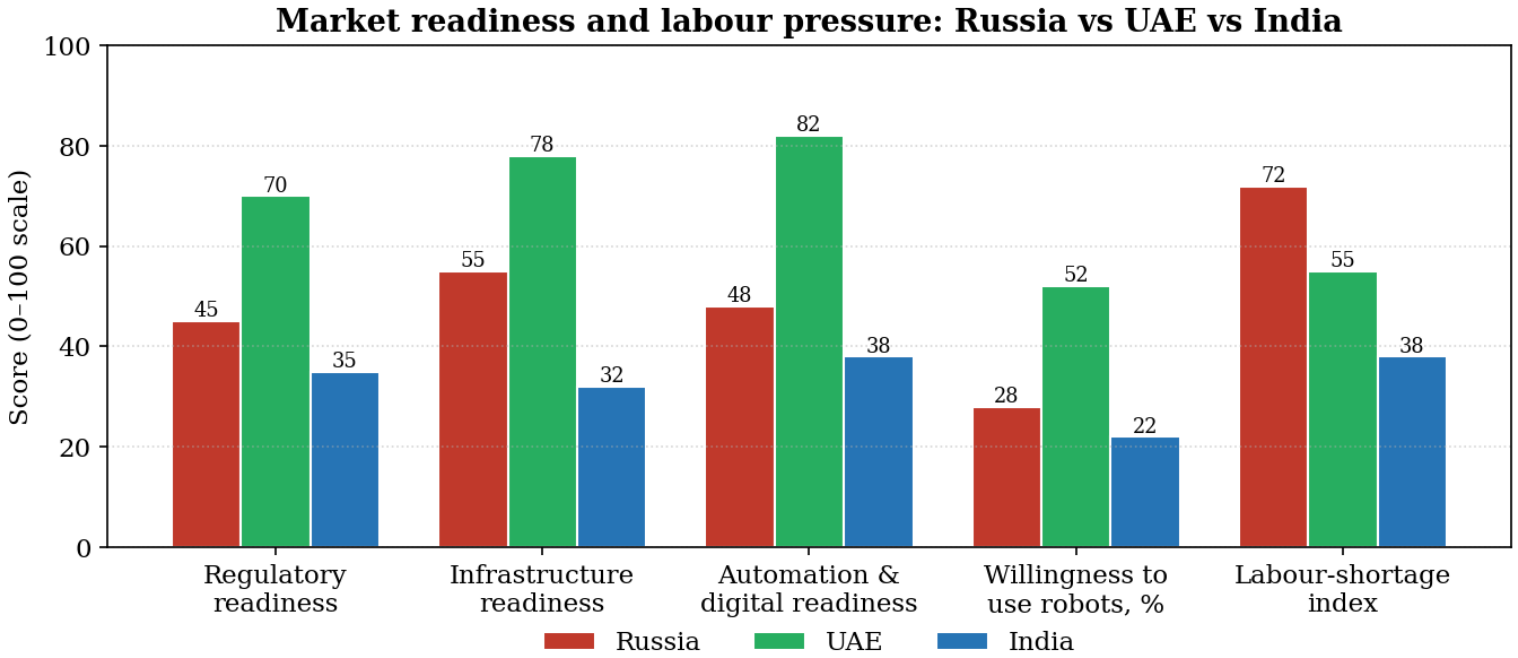
Counterweight — a shortage, not a surplus: automation fills a gap; Moscow needs ~300,000 couriers by 2030.

OUR ANSWER

1 **Augmentation, not replacement**
Robots cover the shifts a structural shortage leaves unstaffed — not jobs taken away.

2 **Reskilling with courier unions**
Co-design retraining and messaging; unions de-risk the backlash (risk #4, 12 pts).

3 **Phased, evidence-gated rollout**
Start where acceptance is highest (UAE 52%); prove safety before any scale-up.



Source: case data package (robotics_market_landscape.csv, consumer_acceptance.csv).

READINESS, 0–100

UAE

leads almost every axis

Russia

lags on readiness, but labour pressure 72 - the engine

India

lowest on every axis except volume

RUSSIA

FIRST

Home pilot - strongest demand, the “last 20 m” niche, EFKO captive retail demand

UAE

SECOND

Commercial scaling - ready regulation (12–24 mo), best infra, acceptance 52%, wealthiest buyers

INDIA

DEFERRED

Cheap labour ($1.2/hr) breaks the economics; social resistance the highest

Internal , B2B sale , **RaaS** → **RaaS wins**

INTERNAL	B2B SALE	RAAS
cost saving only	one-off revenue	recurring revenue
no external revenue	loses control & data	keeps control & data
safe but small	ecosystem moat lost	**ecosystem = the moat**

Economics close **as a service** — for the **right buyers**

COST PER DELIVERY

Human courier — Moscow **$8.5**

Arcus target (in-ecosystem) **~$2**



4–6× lower

- Recurring RaaS: client capex → predictable opex
- EFKO keeps ownership, control & the data moat
- Closes only inside the roboport + postamat ecosystem
- Works only where labour is costly — RU $3.5/hr, UAE $8.5/hr — not India $1.2/hr

WHO WE SERVE — B2B RaaS

Not a hardware sale. Not B2C.

RUSSIA — first

EFKO dark-stores + X5 Group

captive retail demand; the first paying commercial pilot

UAE — second

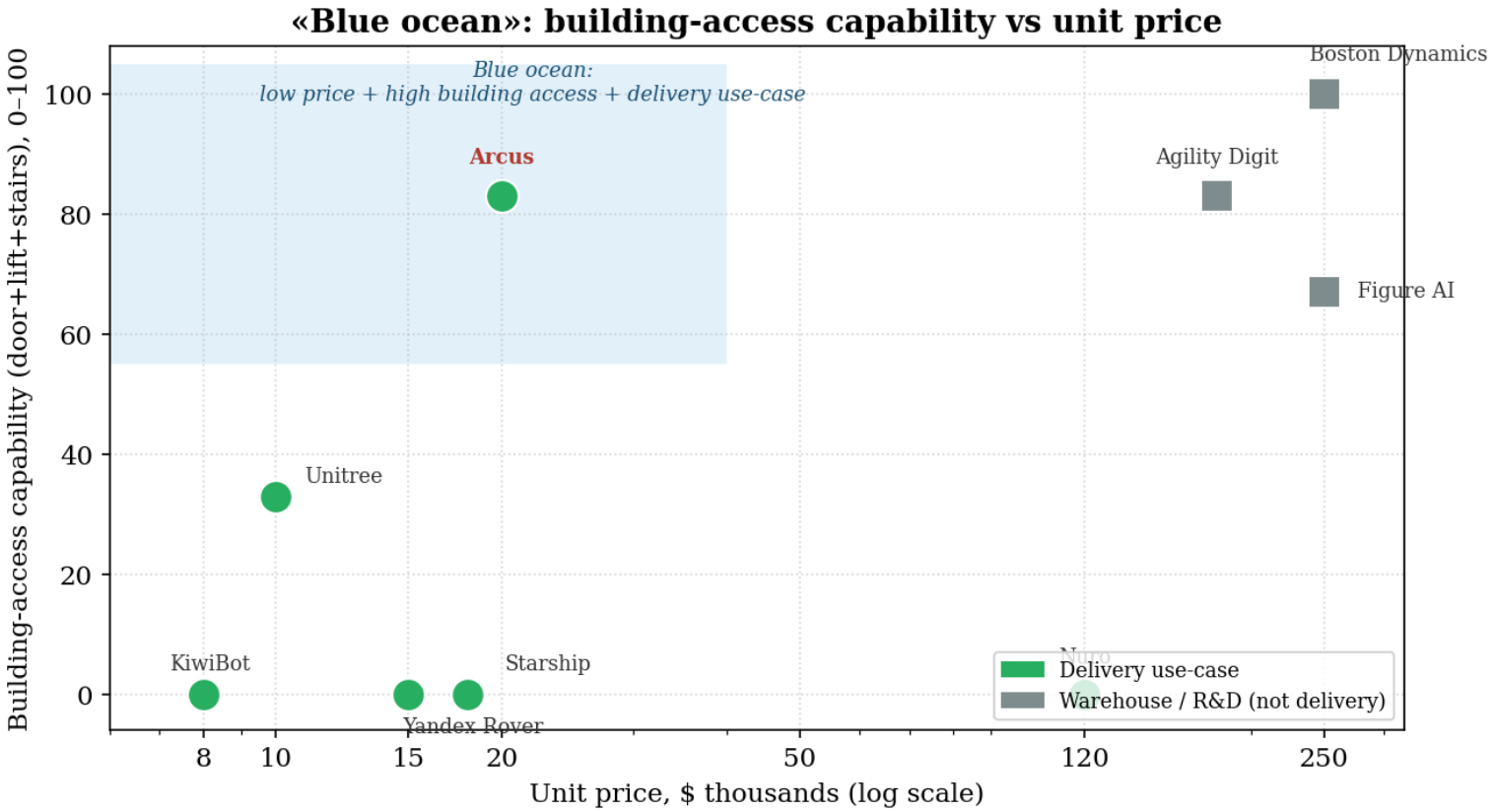
Majid Al Futtaim + Talabat

grocery & last-mile RaaS demand at premium order value

The buyer's pain we remove

Courier cost +15%/yr · 30–45% turnover · ~300,000 shortfall in Moscow by 2030

COMPETITIVE POSITIONING



Source: competitive_landscape_robotics.csv (15 players). Access score: door+lift+stairs capability.

The only humanoid courier with building access at **$20k** - against **$180–250k** warehouse humanoids.

Wheeled rovers stop at the door.
Premium humanoids aren't in delivery.
Arcus owns the empty space.

A humanoid courier can’t launch **alone**

RUSSIA - FIRST

X5 + EFKO dark-stores

paying pilot, national scale

Min. Digital Dev. + Moscow City Hall

legal pilot zone

Courier unions

social trust, de-risk backlash

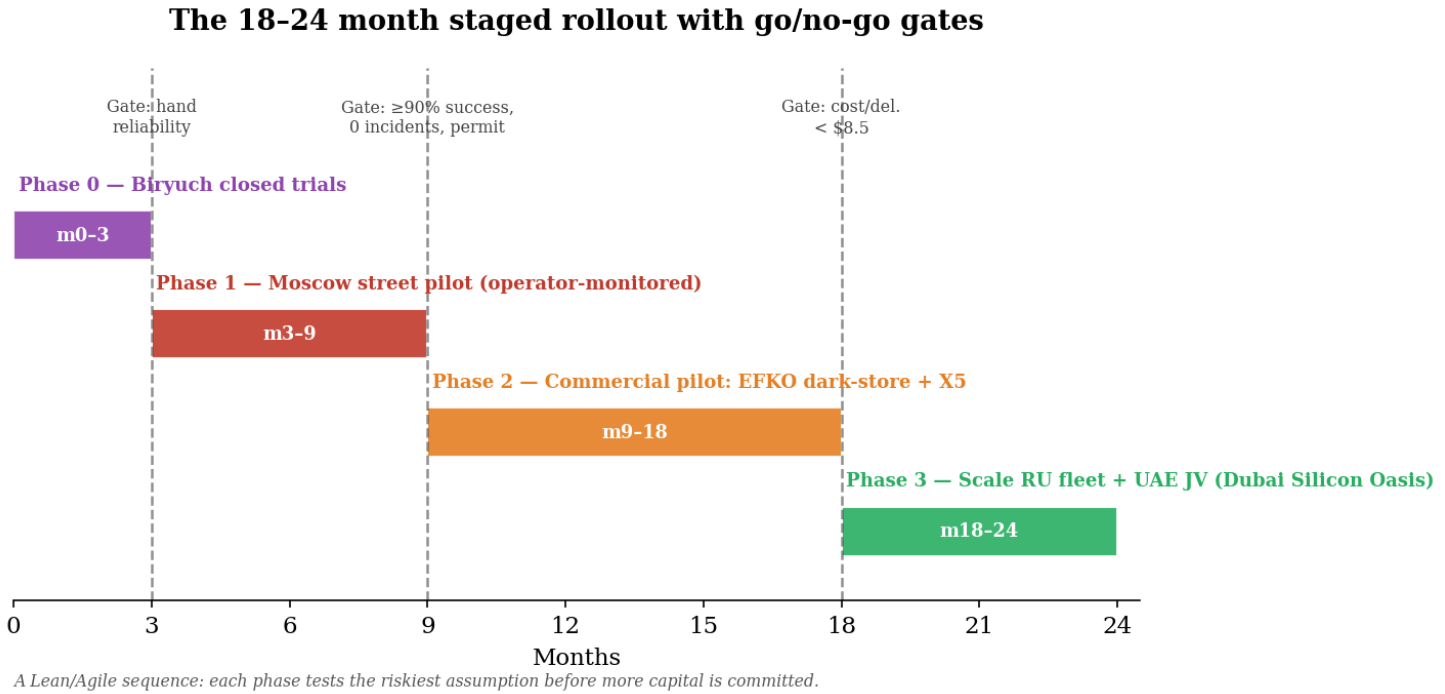
UAE - SECOND

RTA Dubai + Dubai Future Foundation

state pull, AV Strategy 2030

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RaaS grocery & last-mile demand



A Lean/Agile sequence: each phase tests the riskiest assumption before more capital is committed.

Month 18

≥90% delivery success , 0 incidents , cost < $8.5

Month 24

UAE joint venture signed , 10 robots live , economics close

CONCLUSION

Launch Arcus as **RaaS.** Russia first, the UAE next. Win the **last 20 metres.**

SUCCESS

A Moscow route at ≥90% success below the $8.5 benchmark; a UAE JV with 10 robots by month 24.

FAILS IF

Regulation stalls, the hands prove unreliable, or the economics don’t close - so each phase is gated.

Questions & discussion